

Metrics, Error Analysis and Responsible Evaluation

Make model decisions with valid boundaries and decision-aware metrics.



Learning outcomes

- 01  Map metrics to operational costs.
- 02  Analyze meaningful data segments.
- 03  Report uncertainty and failure modes.

The concept system

01

metric choice

02

confidence

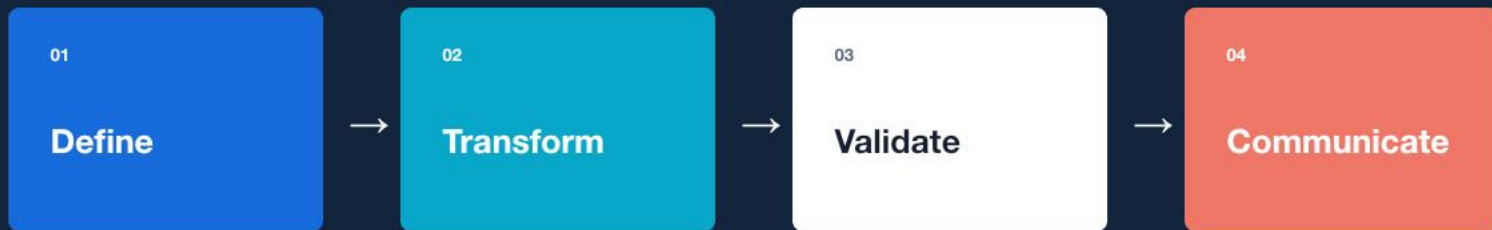
03

segments

04

failure modes

A repeatable reasoning loop



Write an evaluation memo for a high-impact prediction workflow.

Guided lab

LAB BRIEF

Write an evaluation memo for a high-impact prediction workflow.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A decision memo with metrics, slices, and a release recommendation.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Evaluation Decision Memo

Validity 35%, metric reasoning 25%, error analysis 20%, uncertainty 10%, communication 10%.

What should remain after the lesson?

CONCEPT

metric choice + confidence

PRACTICE

Write an evaluation memo for a high-impact prediction workflow.

EVIDENCE

A decision memo with metrics, slices, and a release recommendation.
